GenAI Challenge 2026

From Multilingual NLI to Evidence-Grounded Dialogue Generation

Generative AI | OST - Ostschweizer Fachhochschule | Fall 2026

Student project brief and assessment specification

Why this challenge matters

Imagine that a federal vote is only days away. Your group chat is full of confident claims: “The proposal will definitely cost more.”, “The government says the opposite.”, “I heard the booklet confirms it.” Who is right? And where, exactly, does the official source say so?

In Switzerland, voters receive the *Abstimmungsbüchlein*: the Federal Council’s official explanations of the proposals on the ballot. It contains summaries, background, arguments, recommendations and often legal or financial details. It is designed to inform, but it can still be challenging to navigate, especially when a claim is nuanced, buried in a long PDF, or written in another national language.

Your mission: build an AI fact-checker that can read the booklet, locate evidence and judge whether a claim is supported, contradicted or not addressed. Then turn that same source material into an engaging two-person podcast that someone without specialist knowledge would actually want to listen to. There is one catch: the podcast must fact-check itself.

Meet the source material

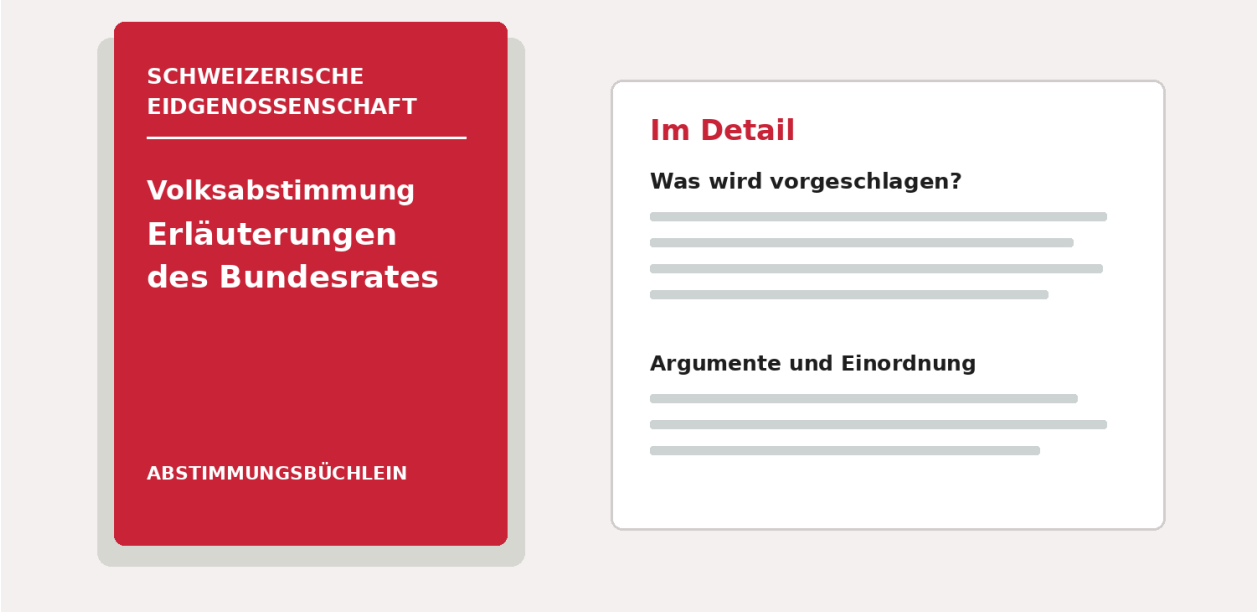


Figure 1: AI-generated illustration of the Swiss federal voting booklet and its explanatory content (not a reproduction of a particular edition).

Two challenges, one trustworthy explanation

Stage 1: Verify. Implement both official NLI tasks: claim versus reference passage, and claim versus complete booklet with supporting evidence. Work across German, French and Italian, including cross-lingual combinations.

Stage 2: Explain. Generate a balanced, accessible 8–12-minute dialogue, submit both script and audio, extract its factual claims, verify them with Stage 1, and revise claims that are not supported.

Challenge overview

Build a two-stage AI application for Swiss federal voting booklets. Stage 1 verifies multilingual claims against supplied reference passages and complete official booklets. Stage 2 generates an accessible two-speaker explanatory dialogue and verifies its factual assertions using the Stage 1 system. The final product includes both a written script and playable audio.

Teams	Core model	Audio	Submission
2–3 students	Apertus v1.5 for NLI	8–12 minutes	Git repository, report, live demo

Important dates

- **Project start:** 23 October 2026
- **Project check-in:** Exercise lesson on 20 or 27 November 2026
- **Live evaluation:** 18 December 2026 (last day of class)
- **Report deadline:** One week before the final written exam (provisional, depends on the exam date)

Administrative rules

- Work in teams of two to three. All members are responsible for understanding the full solution.
- This is a solution-oriented project: the architecture is open, as long as the mandatory outputs and checks work.
- Submit a reproducible Git repository and a short report in English. The submission channel and deadlines will be announced on Moodle.
- Use the official voting booklets as the authoritative source for booklet-grounded inference. Keep political explanations descriptive and balanced.
- **Check-in:** Approach the lecturer or the TA during one of the two exercise lessons. Each team is responsible for initiating its own check-in. The check-in is informal: show your progress, your architecture and your planned features.
- **Live evaluation:** All solutions are presented in a live evaluation on the last day of class. Details will follow on Moodle.

Technical details - Stage 1: NLI

Solve BOTH original HackApertus Track 2A tasks. The inference direction is reference/booklet (premise) to claim (hypothesis). Labels: 0 entailment, 1 neutral, 2 contradiction.

Task	Input	Output
Task B · Beginner / reference NLI	Reference passage + claim	NLI label; preserve case ID
Task A · Advanced / booklet NLI	Booklet PDF + claim	NLI label and verbatim evidence with page number when applicable

- Support German, French and Italian, including cross-lingual pairs.
- For entailment and contradiction, return relevant source quotation(s); for neutral, evidence may be empty.

- Provide a runnable CLI or equivalent adapter for the original challenge input/output format and report tokens and latency.
- Evaluate both tasks separately with macro-F1; discuss evidence quality and representative errors.

Technical details - Stage 2: Dialogue

Select a Swiss voting booklet and generate a two-speaker explanation for a general, non-expert audience. Produce a written transcript and a playable audio file of 10 ± 2 minutes.

- Cover the proposal, background, central arguments for and against, official recommendations, and important caveats.
- Extract factual assertions from the draft dialogue; distinguish them from questions, greetings, and attributed opinions.
- Check each factual assertion against the booklet using the Stage 1 NLI implementation. Record labels and original evidence.
- Revise, qualify, or remove neutral/contradicted claims, then recheck affected claims before finalizing the script and audio.

Minimum Viable Product - passing-grade eligibility (4.0)

- Working reference-text NLI and booklet-level NLI with the three required labels.
- Evidence quotations with page references for booklet-level predictions (scored as HackApertus booklet coverage).
- A complete 8–12-minute, two-speaker script AND playable audio.
- A functioning extraction → NLI verification → revision/recheck workflow.
- A reproducible Git repository, technical report, presentation and Q&A.

Passing-grade eligibility requires completion of all core components. The required criteria award up to 60 points based on quality. Meeting the checklist alone does not guarantee 60 points or a grade of 4.0.

Deliverables

- Mandatory project check-in with the lecturer or TA (see Administrative rules).
- Source code, README, setup instructions, model and dependency details, and runnable commands.
- Stage 1 prediction outputs and evaluation results for both NLI tasks.
- Stage 2 final script (with speaker turns) and audio recording.
- Short English report: introduction, system architecture diagram, implementation, evaluation, limitations, lessons learned, team contributions and an AI usage declaration (which AI tools were used, for what purpose, such as code, writing or debugging, and how their output was checked).
- Live presentation: architecture, demo of both stages, one concrete claim-correction example, results and Q&A.

Evaluation rubric (100 points)

Criterion	Status	Points	Evidence expected
Reference & booklet NLI	**Required**	10	Macro-F1 of the HackApertus evaluation for Reference NLI (Task B) and Booklet NLI (Task A).
Booklet coverage	**Required**	10	HackApertus evidence score (Hit@5) for Booklet NLI (Task A): gold passages found among the first five evidence quotations.
Dialogue & audio	**Required**	10	A complete 8–12-minute, two-speaker script AND playable audio.
Verification workflow	**Required**	15	A functioning extraction → NLI verification → revision/recheck workflow.
Git repository & report	**Required**	5	A reproducible Git repository and technical report, including an AI usage declaration.
Presentation & Q&A	**Required**	10	Working demo, understanding and analysis.
Performance & additional features	Additional	40	Optimisations and features beyond the required deliverables (see below).

Reference & booklet NLI and booklet coverage: both criteria use the official HackApertus scoring, as explained in the Q&A slides in this repository: `HackApertus/Q&A slides/hackapertus-qa.pdf`. Reference & booklet NLI is based on the label-only macro-F1 of Task B and Task A; booklet coverage is based on the evidence score (Hit@5) of Task A. Run `evaluate.py` from the starter repository and report both tasks separately, together with multilingual/cross-lingual analysis and error analysis. The official HackApertus task-specific metrics, output requirements and evidence criteria take precedence over any shorthand description in this document. Stage 2 (verified dialogue and audio) uses the course-specific criteria in the table.

Performance & additional features: points are awarded for work beyond the required deliverables. Performance means general optimisations anywhere in the system, whether in the NLI or the dialogue generation, for example reducing token usage (retrieval, page selection, compression) or inference time (parsing, indexing, caching, batching), with measured before/after numbers in the report. Additional features include, for example, live evidence viewing, an interactive source-grounded chat, or a more fluent and accessible dialogue. More ideas are listed under Optional features / higher-grade extensions.

Grade conversion: $\text{grade} = \min\left(6.0, 1.0 + 5.0 \times \frac{\text{points}}{100}\right)$, subject to course rounding policy. Completing all required deliverables gives 60/100, which corresponds to 4.0; the additional criteria raise the grade above 4.0.

Live evaluation procedure

- Team introduction and architecture overview (approximately 2–3 minutes).
- Demonstration of Stage 1 reference and booklet NLI, including evidence.
- Play an excerpt of the final podcast and inspect the associated transcript and verification log.
- Examiner selects a factual statement and asks the team to show its source and verification outcome.
- Q&A about limitations, errors, design decisions, and team contributions.

Optional features / higher-grade extensions

- Token usage optimisation, in the NLI or the dialogue generation (retrieval, page selection, compression).
- Inference time optimisation, in the NLI or the dialogue generation (parsing, indexing, caching, batching).
- Automated topic-coverage score with evidence of missing or underrepresented booklet sections.
- Story critic for clarity, structure, balance, repetition, and targeted rewriting.
- Audience personas, such as first-time voters or readers with limited technical background.
- Interactive chat interface with source-grounded follow-up answers.
- Multilingual podcast generation and cross-lingual verification.
- Traceability and observability: a claim verification log that links every dialogue statement to its label and evidence.
- Ablation experiments, human evaluation, and comparisons of verification before and after revision.
- Further efforts to make the dialogue coherent, fluent, accessible, and balanced.
- Further efforts to detect bias in the dialogue.
- Quality dimensions: factual reliability • booklet coverage • traceability • fluency • accessibility

System Architecture | Two-Stage Challenge

From official source documents to an evidence-grounded, accessible audio explanation.

INPUT — **Official Swiss voting booklet + claims**
Booklet PDF · vote metadata · reference passages

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STAGE 1 — **Apertus-based NLI**
Task B: claim + reference text → label | Task A: claim + booklet → label + evidence

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STAGE 2 — **Dialogue planning and generation**
Select key topics · represent arguments fairly · write two-speaker script

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CLAIM EXTRACTION + NLI QUALITY GATE
Extract factual assertions → check against source → record label and evidence

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REVISION LOOP
Entailed: retain and cite | Neutral: qualify/revise | Contradicted: correct/remove
Recheck changed claims before finalization

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OUTPUT — **Verified script + audio podcast**
8–12 minutes · two speakers · traceable evidence log

Student resources and supporting documentation

- Official HackApertus Track 2A page: https://hackapertus.notion.site/track-2a-ost
- HackApertus Track 2A Q&A slides (PDF) in this repository at `HackApertus/Q&A slides/hackapertus-qa.pdf`; detailed Task A/Task B, API, evaluation and scoring guidance.
- Course challenge slides (HTML) provided on Moodle; background, examples and NLI workflow.
- Hugging Face benchmark: https://huggingface.co/datasets/OSTswiss/MNLIoverSwissVotingBooklets
- Federal Chancellery booklet archive (original PDFs): https://www.bk.admin.ch/de/sammlung-der-abstimmungsbuechlein-seit-1978

The official HackApertus materials specify Stage 1. This course document additionally specifies the dialogue-generation, audio and verification requirements for Stage 2.